Lecture 4

1/28/26

Lecture overview < Ray tracing / Lens equations | Formalisms for finding image positions, magnification, etc.

Ray Tracing

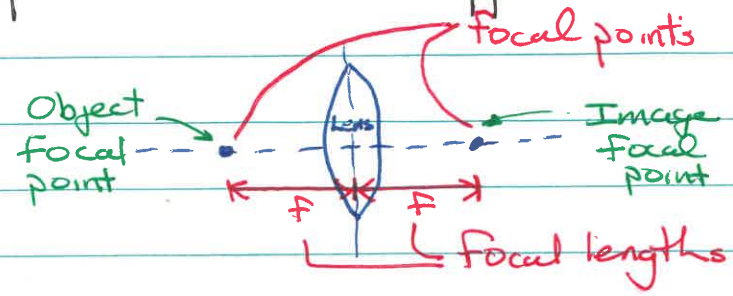
All rays from a given point in object come to focus at same point in image (ignoring aberrations)

Ray tracing uses three particular rays that are easy to deduce & draw to find image location, image size, & magnification. Any two of these rays are sufficient but customary to draw all three as a double check. See Tutorial in textbook.

Look at converging (bulging) lenses.

Small bulge ⇒ less lens power
Large bulge ⇒ more lens power

Lenses have two "focal points" symmetrically positioned at a distance "f," called the "focal length," along the optical axis on opposite sides of the lens



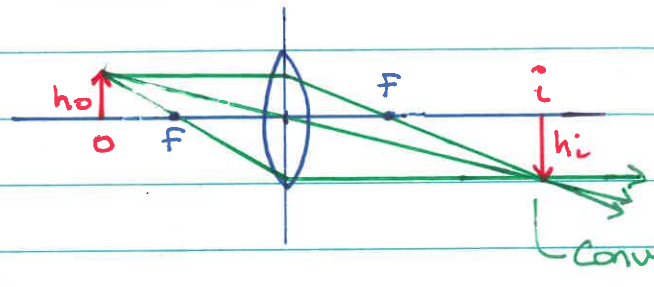


Image = { Real, Inverted, Magnified }

Reviewed ray tracings on PP slide ("Attributes of a Converging Lens") w/ three different object positions.

Decreasing o ⇒ { Increasing i (image distance), Increasing h_i (image size) }

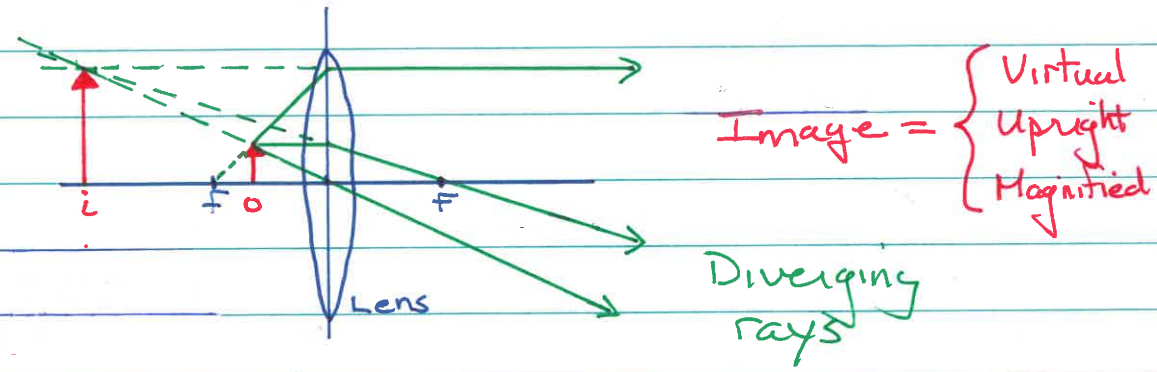
Discussed "real images" & "virtual images"

<u>Real Image</u>:
- light is actually brought to focus
- image has energy to excite camera pixels & burn a piece of paper

Real images are only formed if the object distance, o, is greater than the focal length, f.

<u>Virtual Image</u>:
- light is <u>NOT</u> brought to focus
- no energy at position of image

Virtual images are formed if the object distance is less than the focal length. In this case, the lens is <u>NOT</u> powerful enough to bring the rays to focus. Upshot: <u>NO</u> real image.



Can use another converging lens, like an eye, to bring the diverging rays to focus to form a real image, like on a retina.

The eye interprets the diverging rays as coming from a "virtual image" positioned at a location upstream from the object where there is <u>NO</u> actual light energy.

Always get an upright virtual magnified image when object positioned within focal length of lens.

This class will <u>NOT</u> look at diverging lenses (except in a HW problem).

<u>Lens Equations</u>

Lens equations are a mathematical complement to ray tracing. They give the same answers.

Thin Lens Eq $\boxed{\frac{1}{i} + \frac{1}{o} = \frac{1}{f}}$

HW - prove thin-lens eqn using a similar triangles argument

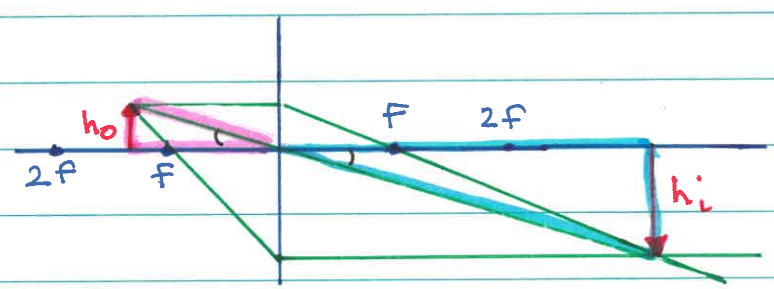
The thin-lens eqn can be used to solve for image position given object position & focal length (among other possibilities, e.g., calculate f given i & o).

Magnification can be defined as a ratio of image height to object height & image position to object position.

$$\text{Magnification} = m = \frac{\text{image height}}{\text{object height}} = \frac{h_i}{h_o} = -\frac{i}{o}$$

Minus sign for inverted image

Thus, a knowledge of i & o from the thin-lens eqn allows calculation of magnification



and in the drawing above are similar triangles, so angles are the same, & ratios of similar side lengths in the two triangles are the same

$$\frac{h_i}{h_o} = \frac{i}{o}$$

The same result can be obtained by noting that the angles opposite the two heights in the two triangles are the same.

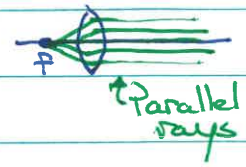
$$\tan\theta = \frac{h_o}{o} = \frac{h_i}{i} \Rightarrow \frac{h_i}{h_o} = \frac{i}{o}$$

Fill out chart showing characteristics of image, exemplary applications, etc. based on object position relative to focal length

Object Distance	Image Type	Image Distance	Image Orientation	Image Size	Application(s)
$\infty > o > 2f$	R	$f < i < 2f$	Inverted	Minified	eye, camera
$o = 2f$	R	$i = 2f$	Inverted	Mag = 1	photocopier
$2f > o > f$	R	$2f < i < \infty$	Inverted	Magnified	μscope objective
$o = f$	Ø	No Focus, Light parallel	No IMAGE	Ø	collimator
$o < f$	V	$\lvert i \rvert > 0$	Upright	Magnified	μscope eyepiece

↑ image to left of object

Collimators use $o = f$ to collimate light



ACTIVITY 1 Virtual Image Formation

See separate Activity page for solution

Name Key

Physics 390 In-Class Activity 1

Virtual Image Formation

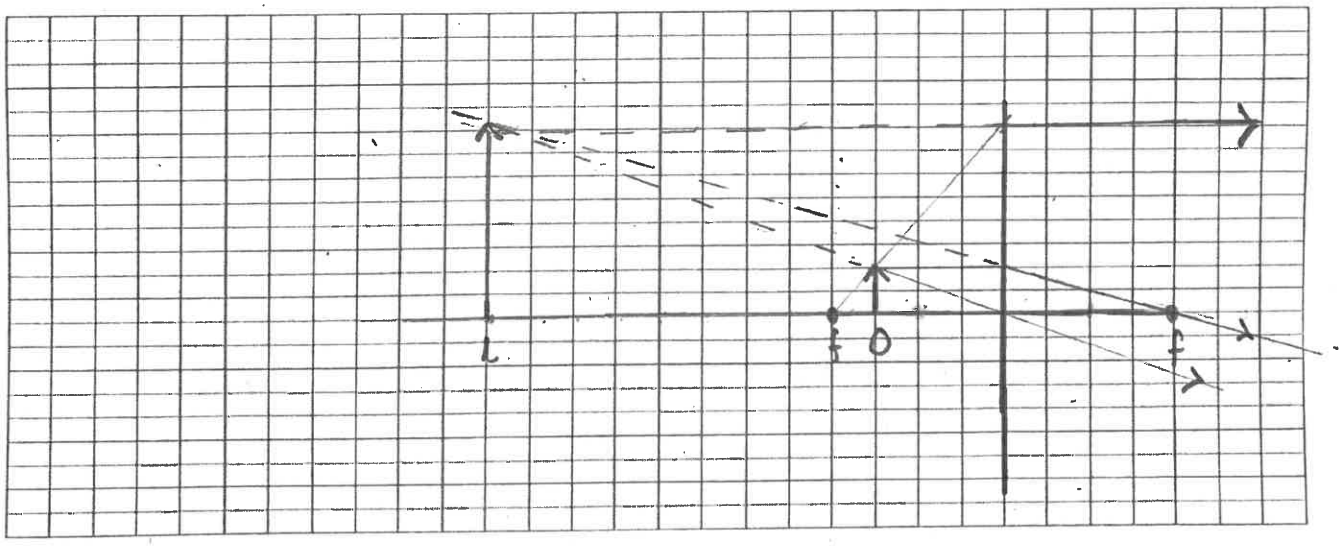
A light bulb is 0.75 m from a converging lens with $f = 1.00$ m.

(a) Compute the image distance and magnification using appropriate equations.

$$1/i + 1/o = 1/f$$

$$1/i = 1/f - 1/o = \frac{o-f}{of}$$

$$\boxed{i = \frac{of}{o-f} = \frac{0.75}{-0.25} = -3\text{m}}$$

$$\boxed{M = -i/o = 3/0.75 = 4}$$

(b) Repeat your analysis by drawing a ray diagram on the graph paper below. Make your object fairly small because part (a) predicts significant magnification.